

MUHAMAD IMRAN BIN KHASRIN

AI Engineer

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SUMMARY

Entry-level AI Engineer with hands-on experience building computer vision pipelines from data annotation and model training (PyTorch, TensorFlow) to deployment on edge platforms like NVIDIA Jetson. Optimized models using ONNX and TensorRT for real-time inference. Seeking to apply this skillset to practical and user-driven applications.

PROJECTS

AI-Powered Workflow Automation Platform | [GitHub](#)

Jan 2026 - Feb 2026

- Designed and developed an end-to-end AI automation platform that converts natural language instructions into executable API workflows using LLM reasoning and semantic retrieval.
- Built a vector database knowledge system that transforms API specifications into embeddings for semantic API retrieval and tool selection.
- Developed an LLM-powered workflow generation engine capable of translating user requests into multi-step API execution plans and structured JSON outputs.
- Built and deployed a full-stack application using FastAPI, React, Docker, MySQL, and Qdrant.
- Implemented automated API matching, validation, and workflow generation to support low-code automation use cases.

Smart Parking IoT Monitoring System (Embedded + Fullstack System) | [GitHub](#)

Apr 2026 - May 2026

- Developed an IoT smart parking system capable of monitoring specific parking slots in real time through embedded sensors and wireless communication.
- Built BLE-based data pipelines and backend APIs that processed parking occupancy updates with less than 2-second refresh latency.
- Developed a real-time dashboard supporting occupancy visualization, filtering, and administrative management.
- Implemented sensor health monitoring with automated offline detection and anomaly tracking to improve system reliability.

Hydrogen Production Capacity Classification Using Machine Learning

Mar 2025 - Jul 2025

- Implemented data preprocessing workflows including missing value handling, feature selection, and dataset transformation.
- Built and benchmarked Random Forest, Gradient Boosting, and SVM classifiers on structured energy and economic datasets.
- Automated hyperparameter tuning using Optuna, GridSearchCV, and RandomizedSearchCV to optimize model performance.
- Evaluated model robustness through multiple train-test split strategies and comprehensive performance metrics analysis.

WORK EXPERIENCE

ROBOMY SDN BHD

Kuala Lumpur, MY

AI Engineer Internship

May 2026 - Present

- Engineered and validated computer vision datasets using CVAT for object detection model training and evaluation.
- Developed and streamlined machine learning pipeline workflows covering data preprocessing, model training, evaluation and optimization for computer vision applications.
- Programmed and executed edge AI inference experiments on Raspberry Pi to evaluate real-time object detection performance in resource-constrained environments.

- Developed an Android application in Kotlin integrating on-device YOLO inference with TensorFlow Lite for real-time object detection.
- Optimized mobile inference using TensorFlow Lite with NNAPI, GPU delegation and CPU fallback to improve real-time performance on Android devices.
- Standardized and documented end-to-end AI workflows from dataset preparation to edge deployment to improve reproducibility and consistency.

PORT KLANG FREE ZONE SDN BHD

Klang, MY

Data Analyst Internship

Sep 2025 - Dec 2025

- Cleaned and analyzed large-scale procurement datasets using Excel and SQL to identify cost-saving opportunities, supplier performance trends, and purchasing inefficiencies.
- Generated weekly and monthly performance reports by translating complex data into actionable insights for supervisors, leading to a 15% improvement in procurement cycle time through data-driven recommendations.
- Collaborated with IT teams to validate data pipelines and ensure system data consistency.
- Worked alongside procurement and finance teams to align data reporting with business goals and improve data collection processes.

PANTAI BHARU HOLDINGS SDN BHD

Klang, MY

Junior Web Developer Internship

Oct 2021 - Mar 2022

- Collaborated with senior developers to build and maintain internal web applications using HTML, CSS, JavaScript, and PHP. Identified and resolved over 20% of frontend bugs in the production system, improving system stability and user experience.
- Developed a cafe food-ordering web app that streamlined the ordering process, reducing wait times by 25%.
- Built a customer information portal for the Proton department using PHP and MySQL, enabling efficient data entry and retrieval.

EDUCATION

UNIVERSITI TEKNOLOGI MARA

Shah Alam, MY

Bachelor Of Information Systems (Hons.) Intelligent Systems Engineering

Oct 2022 - Mar 2026

CGPA: 3.48

Relevant Coursework: Machine Learning, Data Mining, Time Series Analysis and Forecasting, Optimization Algorithms and Applications, Fuzzy Logic, Programming For Data Science

POLITEKNIK SULTAN HAJI AHMAD SHAH (POLISAS)

Kuantan, MY

Diploma In Business Information Systems

Dec 2018 - Sep 2022

CGPA: 3.68

Relevant Coursework: Web development, Web Programming, Database Management System, Project Management, Digital Entrepreneurship Development

PERSONAL SKILLS

- Programming Languages: Experienced in Python, R, C++, Java, HTML, CSS, JavaScript, Typescript, PHP, SQL, Matlab
- Data Visualization: Power BI, Tableau, Matplotlib, Seaborn
- Deep Learning Frameworks: TensorFlow, PyTorch, Keras
- Web Frameworks and Database: React, Flask, Vue.js

ACHIEVEMENTS

- Dean's List (Semester 3–6), UiTM
- Dean's List (Semester 1–4), POLISAS
- Special Industrial Award from MKZ Global Resources